

Electrodynamics

Potentials, gauge freedom and multipole expansion

study notes

1 Why potentials? six unknowns become four

If we know the **divergence and curl** of a vector field, we have its complete information. So to find the unknown fields $\vec{E}$ and $\vec{B}$ we rewrite them in terms of potentials. In 3D, $\vec{E}$ and $\vec{B}$ together give **6 unknowns**; written through the scalar potential V and vector potential $\vec{A}$ they give only **4 unknowns**.

Maxwell's equations

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (1)$$

$$\nabla \cdot \vec{B} = 0 \quad \text{no net magnetic charge density} \quad (2)$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (3)$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad (4)$$

2 Defining $\vec{A}$ and V from the homogeneous equations

Since $\nabla \cdot \vec{B} = 0$, write $\vec{B}$ as a curl of the **vector potential** $\vec{A}$:

$$\vec{B} = \nabla \times \vec{A} \quad (5)$$

Put this in Faraday's law (3):

$$\begin{aligned} \nabla \times \vec{E} &= -\frac{\partial}{\partial t} (\nabla \times \vec{A}) = -\left(\nabla \times \frac{\partial \vec{A}}{\partial t}\right) \\ \Rightarrow \nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t}\right) &= 0 \end{aligned}$$

Because the curl of this vector is zero, we can write it as the gradient of a scalar.

$$\Rightarrow \vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla V$$

Fields from potentials

$$\begin{aligned} \vec{B} &= \nabla \times \vec{A} \\ \vec{E} &= -\nabla V - \frac{\partial \vec{A}}{\partial t} \end{aligned} \quad (6)$$

Eqs. (5) and (6) (the second is the boxed pair above) express the magnetic and electric fields in terms of the potentials.

3 Equations for the potentials two coupled inhomogeneous equations

Gauss's law (1)

$$\begin{aligned}\nabla \cdot \left(-\nabla V - \frac{\partial \vec{A}}{\partial t} \right) &= \frac{\rho}{\epsilon_0} \\ \Rightarrow -\nabla^2 V - \frac{\partial}{\partial t} (\nabla \cdot \vec{A}) &= \frac{\rho}{\epsilon_0} \\ \Rightarrow \nabla^2 V + \frac{\partial}{\partial t} (\nabla \cdot \vec{A}) &= -\frac{\rho}{\epsilon_0}\end{aligned}\tag{7}$$

Ampère–Maxwell law (4)

Insert $\vec{B} = \nabla \times \vec{A}$ and $\vec{E}$ from (6):

$$\begin{aligned}\nabla \times (\nabla \times \vec{A}) &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(-\nabla V - \frac{\partial \vec{A}}{\partial t} \right) \\ \text{use } \nabla \times (\nabla \times \vec{A}) &= \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} \quad (\text{vector identity}) \\ \Rightarrow \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} &= \mu_0 \vec{J} - \mu_0 \epsilon_0 \nabla \frac{\partial V}{\partial t} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} \\ \Rightarrow \left(\nabla^2 \vec{A} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} \right) - \nabla \left(\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) &= -\mu_0 \vec{J}\end{aligned}\tag{8}$$

Eqs. (7) and (8) are **inhomogeneous second-order differential equations** and must be solved *simultaneously*.

4 Gauge freedom gauge degrees of freedom

Two different sets $(V, \vec{A}) \rightarrow (\vec{E}, \vec{B})$ and $(V', \vec{A}') \rightarrow (\vec{E}, \vec{B})$ can describe the same physical fields. If $\vec{A}$ and V are the magnetic vector and electric scalar potentials, consider a change

$$V \rightarrow V' = V + \beta, \quad \vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\alpha}.$$

In going to potentials we managed to reduce the number of variables from 6 to 4, but the defining equations (5) and (6) do *not* uniquely fix the potentials. We are free to impose an extra condition on V and $\vec{A}$ as long as nothing happens to $\vec{E}$ and $\vec{B}$. This is the **gauge freedom**.

(a) $\vec{B}$ must not change

$$\begin{aligned}\vec{B} &= \nabla \times \vec{A} \quad \text{and} \quad \vec{B} = \nabla \times \vec{A}' \\ \Rightarrow \nabla \times \vec{A} &= \nabla \times (\vec{A} + \vec{\alpha}) = \nabla \times \vec{A} + \nabla \times \vec{\alpha} \\ \Rightarrow \nabla \times \vec{\alpha} &= 0 \\ \Rightarrow \vec{\alpha} &= \nabla \lambda \quad (\text{a})\end{aligned}$$

(b) $\vec{E}$ must not change

$$\begin{aligned}
 \vec{E} &= -\nabla V - \frac{\partial \vec{A}}{\partial t} \quad \text{and} \quad \vec{E} = -\nabla V' - \frac{\partial \vec{A}'}{\partial t} \\
 \Rightarrow -\nabla V - \frac{\partial \vec{A}}{\partial t} &= -\nabla V' - \frac{\partial \vec{A}'}{\partial t} \\
 \Rightarrow -\nabla V - \frac{\partial \vec{A}}{\partial t} &= -\nabla V - \nabla \beta - \frac{\partial \vec{A}}{\partial t} - \frac{\partial \vec{\alpha}}{\partial t} \\
 \Rightarrow \nabla \beta &= -\frac{\partial}{\partial t} \nabla \lambda \Rightarrow \nabla \left(\beta + \frac{\partial \lambda}{\partial t} \right) = 0 \\
 \Rightarrow \beta + \frac{\partial \lambda}{\partial t} &= k(t) \\
 \Rightarrow \beta &= -\frac{\partial \lambda}{\partial t} + k(t) \quad (b)
 \end{aligned}$$

Absorb $k(t)$ into λ by redefining $\lambda \rightarrow \lambda - \int_0^t k(t') dt'$. Then

$$\beta = -\frac{\partial \lambda}{\partial t} \quad (c)$$

Gauge transformation

$$V' = V - \frac{\partial \lambda}{\partial t}, \quad \vec{A}' = \vec{A} + \nabla \lambda$$

Here $\lambda(\vec{r}, t)$ is any scalar function. Both $\vec{E}$ and $\vec{B}$ are unchanged, so we may use this freedom to simplify (7) and (8).

5 Coulomb gauge $\nabla \cdot \vec{A} = 0$

Choose the divergence of $\vec{A}$ to be zero. Then Eq. (7) reduces to

Poisson's equation

$$\nabla \cdot \vec{A} = 0 \quad \Rightarrow \quad \nabla^2 V = -\frac{\rho}{\epsilon_0}$$

With boundary conditions $V \rightarrow 0$ as $r \rightarrow \infty$:

$$V(\vec{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}', t)}{\mathcal{R}} d\tau' \quad \text{with } \mathcal{R} = \vec{r} - \vec{r}'$$

Meaning. The scalar potential in the Coulomb gauge is determined by the distribution of charge *at the present time*. This implies a signal propagating faster than the speed of light in vacuum, which relativity forbids as a physical signal. (The conflict is only apparent: V alone is not measurable, and the fields $\vec{E}, \vec{B}$ still behave causally.)

Advantages

(i) V is very simple to calculate; it takes the familiar electrostatic form.

Disadvantages

(i) $\vec{A}$ is difficult to calculate.

6 Lorenz (Lorentz) gauge $\nabla \cdot \vec{A} = -\mu_0 \epsilon_0 \partial V / \partial t$

Choose

$$\nabla \cdot \vec{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}.$$

The coupling term in (8) drops out, and (7) loses its $\vec{A}$ dependence:

$$\nabla^2 V - \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon_0}$$

$$\nabla^2 \vec{A} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J}$$

Define the **d'Alembertian** $\square^2 \equiv \nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2}$:

Lorenz gauge

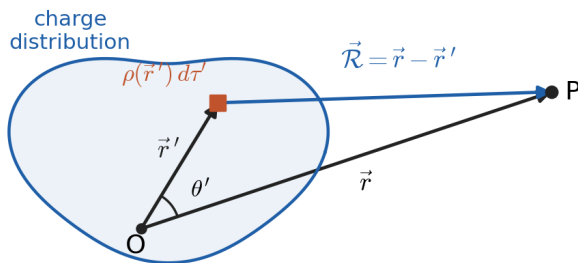
$$\square^2 V = -\frac{\rho}{\epsilon_0} \quad ; \quad \square^2 \vec{A} = -\mu_0 \vec{J}$$

In the Lorenz gauge V and $\vec{A}$ satisfy the **same inhomogeneous wave equation, with a source term on the right**. (Their solutions are the retarded potentials: the same integrals as before, but with ρ and $\vec{J}$ evaluated at the retarded time $t - \mathcal{R}/c$, which respects causality.)

	Coulomb gauge	Lorenz gauge
condition	$\nabla \cdot \vec{A} = 0$	$\nabla \cdot \vec{A} = -\mu_0 \epsilon_0 \partial V / \partial t$
equation for V	$\nabla^2 V = -\rho / \epsilon_0$ (Poisson)	$\square^2 V = -\rho / \epsilon_0$
equation for $\vec{A}$	coupled to V , hard	$\square^2 \vec{A} = -\mu_0 \vec{J}$

7 Multipole expansion of the scalar potential charge distribution, large r

Multipole expansion is the systematic expansion of the potential of an arbitrary *localised* charge distribution in powers of $1/r$. Given some charge and current distribution we want $\vec{E}$ and $\vec{B}$; in terms of potentials we need only V and $\vec{A}$.



O origin of the coordinate system

P observation point where the potential is measured

$\rho(\vec{r}')$ elementary charge distribution at $\vec{r}'$

$\vec{R} = \vec{r} - \vec{r}'$, $\theta' =$ angle between $\vec{r}$ and $\vec{r}'$

We consider an elementary volume $d\tau'$ around $\vec{r}'$, so that the charge in it is uniform. Its total charge is

$$dq = \rho(\vec{r}') d\tau'$$

and, as for a point charge $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$, the potential at $\vec{r}$ due to it is $\frac{1}{4\pi\epsilon_0} \frac{\rho(\vec{r}') d\tau'}{\mathcal{R}}$. Summing over the distribution:

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{\mathcal{R}} d\tau' \quad (9)$$

Step 1 — expand $1/\mathcal{R}$

From the triangle law of vector addition:

$$\mathcal{R}^2 = r^2 + r'^2 - 2rr' \cos \theta'$$

$$\Rightarrow \mathcal{R} = \sqrt{r^2 + r'^2 + (-2rr' \cos \theta')}$$

$$\Rightarrow \mathcal{R}^2 = r^2 \left[1 + \left(\frac{r'}{r} \right)^2 - 2 \left(\frac{r'}{r} \right) \cos \theta' \right] = r^2 (1 + \epsilon)$$

$$\text{where } \epsilon = \frac{r'}{r} \left(\frac{r'}{r} - 2 \cos \theta' \right) \quad (10)$$

$$\therefore \frac{1}{\mathcal{R}} = \frac{1}{r} (1 + \epsilon)^{-1/2} \quad (11)$$

For points well outside the charge distribution $\epsilon \ll 1$, so we may use the **binomial expansion**:

$$(1 + \epsilon)^{-1/2} = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \dots \quad (12)$$

Putting (12) in (11), with powers of ϵ written out ($x \equiv r'/r, c \equiv \cos \theta'$):

$$\epsilon = x^2 - 2xc$$

$$\epsilon^2 = x^2 \left(\frac{r'}{r} - 2c \right)^2 = x^4 - 4x^3c + 4x^2c^2$$

$$\epsilon^3 = x^3 \left(\frac{r'}{r} - 2c \right)^3 = x^6 - 6x^5c + 12x^4c^2 - 8x^3c^3$$

Collect the terms power by power in $x = r'/r$:

$$\frac{r}{\mathcal{R}} = 1 - \frac{1}{2}(x^2 - 2xc) + \frac{3}{8}(4x^2c^2 - 4x^3c + \dots) - \frac{5}{16}(-8x^3c^3 + \dots)$$

$$= 1 + xc + x^2 \left(\frac{3}{2}c^2 - \frac{1}{2} \right) + x^3 \left(\frac{5}{2}c^3 - \frac{3}{2}c \right) + \dots$$

$$\frac{1}{\mathcal{R}} = \frac{1}{r} \left[1 + \left(\frac{r'}{r} \right) \cos \theta' + \left(\frac{r'}{r} \right)^2 \left(\frac{3}{2} \cos^2 \theta' - \frac{1}{2} \right) + \left(\frac{r'}{r} \right)^3 \left(\frac{5}{2} \cos^3 \theta' - \frac{3}{2} \cos \theta' \right) + \dots \right] \quad (13)$$

Step 2 — the coefficients are Legendre polynomials

Equation (13) can be written with Legendre polynomials $P_n(x)$, $n = 0, 1, 2, \dots$:

$$P_0(x) = 1$$

$$P_0(\cos \theta') = 1$$

$$P_1(x) = x$$

$$P_1(\cos \theta') = \cos \theta'$$

$$P_2(x) = \frac{3x^2 - 1}{2}$$

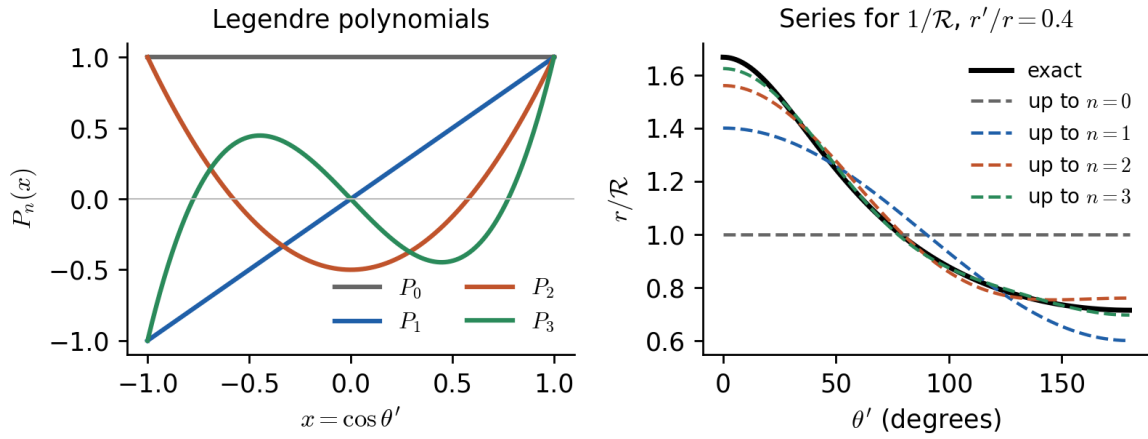
$$P_2(\cos \theta') = \frac{3}{2} \cos^2 \theta' - \frac{1}{2}$$

$$P_3(x) = \frac{5x^3 - 3x}{2}$$

$$P_3(\cos \theta') = \frac{5}{2} \cos^3 \theta' - \frac{3}{2} \cos \theta'$$

Generating function of Legendre polynomials

$$\frac{1}{\mathcal{R}} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \theta') \quad (14)$$



Left: the first four Legendre polynomials. Right: partial sums of (14) against the exact $r/\mathcal{R}$ for $r'/r = 0.4$ — each added term improves the fit, which is why the expansion works for $r \gg r'$.

Step 3 — put the series in the potential

Insert (14) in (9):

$$\begin{aligned}
 V &= \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{\mathcal{R}} d\tau' \\
 \Rightarrow V &= \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \int \frac{(r')^n \rho(\vec{r}') P_n(\cos \theta')}{r^{n+1}} d\tau' \\
 \Rightarrow V &= \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n \rho(\vec{r}') P_n(\cos \theta') d\tau'
 \end{aligned}$$

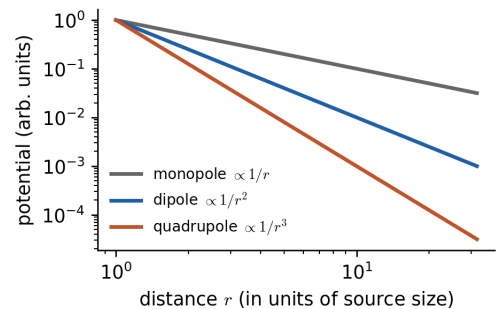
Multipole expansion of V

$$\begin{aligned}
 V &= \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r} \int \rho(\vec{r}') d\tau'}_{\text{monopole}} + \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int \rho(\vec{r}') r' \cos \theta' d\tau'}_{\text{dipole}} \\
 &+ \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \int \rho(\vec{r}') (r')^2 \left(\frac{3}{2} \cos^2 \theta' - \frac{1}{2} \right) d\tau'}_{\text{quadrupole}} \\
 &+ \frac{1}{4\pi\epsilon_0} \frac{1}{r^4} \int \rho(\vec{r}') (r')^3 \left(\frac{5}{2} \cos^3 \theta' - \frac{3}{2} \cos \theta' \right) d\tau' + \dots
 \end{aligned} \tag{15}$$

What each term means

- 1st term — **monopole** term, $V \propto 1/r$.
- 2nd term — **dipole** term, $V \propto 1/r^2$.
- 3rd term — **quadrupole** term, $V \propto 1/r^3$.

At large distances the monopole term goes as $1/r$, so for large r the expansion is ordinarily **dominated by the monopole term**.



- For a point charge at the origin, V_{monopole} is the *exact* potential everywhere.
- If the total charge is zero, then $V_{\text{monopole}} = 0$.

- In a multipole expansion the **choice of origin is important**. If the point charge is not at the origin, it is no longer a pure monopole.
- Ordinarily the dipole moment changes when we shift the origin; but if the **total charge is zero**, the dipole moment is independent of the choice of origin.

Why: shifting the origin by $\vec{a}$ gives $\vec{p}' = \int (\vec{r}' - \vec{a}) \rho d\tau' = \vec{p} - Q\vec{a}$, which equals $\vec{p}$ when $Q = 0$.

Dipole term, explicitly

$$\begin{aligned} V_{\text{dipole}} &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{r^2} \int \rho(\vec{r}') r' \cos \theta' d\tau' \\ &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{r^2} \int \rho(\vec{r}') r' (\hat{r} \cdot \hat{r}') d\tau' \\ &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{r^2} \int \hat{r} \cdot \vec{r}' \underbrace{\rho(\vec{r}') d\tau'}_{dq} \end{aligned}$$

Dipole potential

$$V_{\text{dip}} = \frac{1}{4\pi\epsilon_0} \frac{\hat{r} \cdot \vec{p}}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\vec{r} \cdot \vec{p}}{r^3}, \quad \vec{p} = \int \vec{r}' \rho(\vec{r}') d\tau'$$

$\vec{p}$ is the **dipole moment** of the charge distribution.

Quadrupole term, explicitly

$$\begin{aligned} V_{\text{quad}} &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2r^3} \int (r')^2 (3 \cos^2 \theta' - 1) \rho(\vec{r}') d\tau' \\ &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2r^3} \int (r')^2 [3(\hat{r} \cdot \hat{r}')^2 - 1] \rho(\vec{r}') d\tau' \\ &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2r^3} \int [3(\vec{r}' \cdot \hat{r})^2 - (r')^2] \rho(\vec{r}') d\tau' \end{aligned}$$

Write $\vec{r}' \cdot \hat{r} = \sum_i r'_i \hat{r}_i$ and $(r')^2 = \sum_{i,j} r'_i r'_j \delta_{ij}$, where r'_i are the Cartesian components (x', y', z') of $\vec{r}'$ and δ_{ij} is the Kronecker delta:

$$\begin{aligned} V_{\text{quad}} &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2r^3} \int \left[3 \sum_{i,j} r'_i \hat{r}_i r'_j \hat{r}_j - (r')^2 \right] \rho(\vec{r}') d\tau' \\ &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2r^3} \sum_{i,j} \hat{r}_i \hat{r}_j \int [3r'_i r'_j - (r')^2 \delta_{ij}] \rho(\vec{r}') d\tau' \end{aligned}$$

Define

Quadrupole moment

$$Q_{ij} = \int (3r'_i r'_j - (r')^2 \delta_{ij}) \rho(\vec{r}') d\tau' \quad \implies \quad V_{\text{quad}} = \frac{1}{4\pi\epsilon_0} \sum_{i,j} \frac{Q_{ij} \hat{r}_i \hat{r}_j}{2r^3}$$

So the **monopole moment is a scalar**, the **dipole moment is a vector**, and the **quadrupole moment is a 2nd-rank tensor**.

Topics noted for revision: (1) redo the multipole expansion in Cartesian coordinates; (2) multipole expansion of the energy of a charge distribution in an external field (next section).

8 Charge distribution in an external field energy expansion

A localised charge distribution $\rho(\vec{x})$ placed in an external potential $\phi(\vec{x})$ has electrostatic energy

$$W = \int \phi(\vec{x}) \rho(\vec{x}) d^3x \quad (16)$$

Assume $\phi(\vec{x})$ is *slowly varying* over the region where ρ is non-negligible. Then expand ϕ in a Taylor series about a suitably chosen origin:

$$\phi(\vec{x}) = \phi(0) + \vec{x} \cdot \nabla \phi(0) + \frac{1}{2} \sum_{i,j} x_i x_j \frac{\partial^2 \phi(0)}{\partial x_i \partial x_j} + \dots$$

With $\vec{E} = -\nabla \phi$, i.e. $(\nabla \phi)_i = \partial \phi / \partial x_i$ and $E_j = -(\nabla \phi)_j$:

$$\phi(\vec{x}) = \phi(0) - \vec{x} \cdot \vec{E}(0) - \frac{1}{2} \sum_i \sum_j x_i x_j \frac{\partial E_j(0)}{\partial x_i} + \dots$$

Since $\nabla \cdot \vec{E} = 0$ for an external field (no sources there), we can add $\frac{1}{6} r^2 \nabla \cdot \vec{E}(0) = 0$:

$$\phi(\vec{x}) = \phi(0) - \vec{x} \cdot \vec{E}(0) - \frac{1}{6} \sum_{i,j} (3x_i x_j - r^2 \delta_{ij}) \frac{\partial E_j(0)}{\partial x_i} + \dots \quad (17)$$

Insert (17) in (16):

$$W = \phi(0) \int \rho d^3x - \int \rho \vec{x} d^3x \cdot \vec{E}(0) - \frac{1}{6} \sum_{i,j} \int \rho (3x_i x_j - r^2 \delta_{ij}) d^3x \frac{\partial E_j(0)}{\partial x_i} + \dots$$

Using the definitions of the total charge, dipole moment and quadrupole moment:

Energy in an external field

$$W = Q \phi(0) - \vec{p} \cdot \vec{E}(0) - \frac{1}{6} \sum_{i,j} Q_{ij} \frac{\partial E_j(0)}{\partial x_i} + \dots$$

The expansion shows how the multipoles interact with the external field: the **charge interacts with** $\phi(0)$, the **dipole with** $\vec{E}(0)$, and the **quadrupole with** $\nabla \vec{E}(0)$.

In nuclear physics the quadrupole interaction is of particular interest. Atomic nuclei can possess electric quadrupole moments, and their magnitudes and signs reflect the nature of the forces between neutrons and protons and the shape of the nuclei themselves.

9 Multipole expansion of the vector potential magnetic case

Choosing the gauge

We know

$$\nabla \cdot \vec{B} = 0 \quad (18)$$

$$\vec{B} = \nabla \times \vec{A} \quad (19)$$

and by Ampère's law $\nabla \times \vec{B} = \mu_0 \vec{J}$, so

$$\nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} \Rightarrow \nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J} \quad (20)$$

For comparison, for the scalar potential $\nabla^2 V = -\rho/\epsilon_0$, $\rho \rightarrow 0$ at $r \rightarrow \infty$, and $V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{\mathcal{R}} d\tau'$.

We can add $\nabla \lambda$ to $\vec{A}$ so that $\nabla \times (\vec{A} + \nabla \lambda) = \vec{B}$, because **the curl of a gradient is zero**: we may add a magnetic potential of any function whose curl vanishes. We exploit this freedom to make $\nabla \cdot \vec{A} = 0$:

$$\vec{A}' = \vec{A} + \nabla \lambda \quad (21)$$

$$\nabla \cdot \vec{A}' = \nabla \cdot \vec{A} + \nabla^2 \lambda \stackrel{!}{=} 0$$

$$\Rightarrow \nabla^2 \lambda = -\nabla \cdot \vec{A} \quad (22)$$

If $\nabla \cdot \vec{A} \rightarrow 0$ at infinity, then, by analogy with Poisson's equation,

$$\lambda = \frac{1}{4\pi} \int \frac{\nabla \cdot \vec{A}}{\mathcal{R}} d\tau' \quad (23)$$

So for a particular choice of gauge we can always reduce (20) to

Choosing $\nabla \cdot \vec{A} = 0$ (Coulomb gauge)

$$\nabla^2 \vec{A} = -\mu_0 \vec{J} \quad \implies \quad \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{\mathcal{R}} d\tau' \quad (\vec{J} \rightarrow 0 \text{ at } \infty)$$

The definition $\vec{B} = \nabla \times \vec{A}$ specifies the curl of $\vec{A}$ but says nothing about its divergence, so we are free to choose it according to our need. Zero is the simplest and most natural choice in most cases.

Expansion for a current loop

We want the vector potential of a **localised current distribution at a large distance**: a loop carrying current I , with $d\vec{l}'$ at $\vec{r}'$, observation point P at $\vec{r}$, $\vec{\mathcal{R}} = \vec{r} - \vec{r}'$, angle θ' . Same geometry as for V , so the same expansion of $1/\mathcal{R}$ applies:

$$\begin{aligned} \mathcal{R}^2 &= r^2 + r'^2 - 2rr' \cos \theta' \\ \implies \mathcal{R} &= r[1 + \epsilon]^{1/2}, \\ \epsilon &= \frac{r'}{r} \left(\frac{r'}{r} - 2 \cos \theta' \right) \end{aligned}$$

$$\frac{1}{\mathcal{R}} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \theta') \quad (24)$$

The coefficients of r'/r are the Legendre polynomials; $1/\mathcal{R}$ is the *generating function* for the Legendre polynomials. Then

$$\begin{aligned} \vec{A}(\vec{r}) &= \frac{\mu_0 I}{4\pi} \oint \frac{d\vec{l}'}{\mathcal{R}} \\ &= \frac{\mu_0 I}{4\pi} \oint d\vec{l}' \left[\frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \theta') \right] \end{aligned} \quad (25)$$

Here $\vec{r}(x, y, z)$ and $\vec{r}'(x', y', z')$; the integration is over $\vec{r}'$, so $1/r$ comes outside:

$$\vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \left(\frac{1}{r} \right)^{n+1} \oint d\vec{l}' (r')^n P_n(\cos \theta')$$

Using $P_0 = 1$, $P_1 = \cos \theta'$, $P_2 = \frac{3}{2} \cos^2 \theta' - \frac{1}{2}$:

Multipole expansion of $\vec{A}$

$$\begin{aligned} \vec{A}(\vec{r}) &= \frac{\mu_0 I}{4\pi} \frac{1}{r} \oint d\vec{l}' P_0 + \frac{\mu_0 I}{4\pi} \frac{1}{r^2} \oint d\vec{l}' r' P_1(\cos \theta') + \frac{\mu_0 I}{4\pi} \frac{1}{r^3} \oint d\vec{l}' r'^2 P_2(\cos \theta') + \dots \\ &= \underbrace{\frac{\mu_0 I}{4\pi} \frac{1}{r} \oint d\vec{l}'}_{\text{monopole}} + \underbrace{\frac{\mu_0 I}{4\pi} \frac{1}{r^2} \oint r' \cos \theta' d\vec{l}'}_{\text{dipole}} + \underbrace{\frac{\mu_0 I}{4\pi} \frac{1}{r^3} \oint r'^2 \left(\frac{3}{2} \cos^2 \theta' - \frac{1}{2} \right) d\vec{l}'}_{\text{quadrupole}} + \dots \end{aligned} \quad (26)$$

The 1st term is the monopole ($\propto 1/r$), the 2nd the dipole ($\propto 1/r^2$), the 3rd the quadrupole ($\propto 1/r^3$), and so on.

The magnetic monopole term vanishes

The total vector displacement around a closed loop is zero:

$$\oint d\vec{l}' = 0$$

No magnetic monopole

$\therefore$ the magnetic monopole term is always zero.

This is why we do not see magnetic monopoles in nature. In the absence of a monopole term, **the dominant term is the dipole**:

$$\begin{aligned} \vec{A}_{\text{dip}}(\vec{r}) &= \frac{\mu_0 I}{4\pi} \frac{1}{r^2} \oint r' \cos \theta' d\vec{l}' \\ &= \frac{\mu_0 I}{4\pi r^2} \oint (\hat{r} \cdot \vec{r}') d\vec{l}' \quad \left[\because \hat{r} \cdot \vec{r}' = r' \cos \theta' \right] \end{aligned} \quad (27)$$

Written as an area integral,

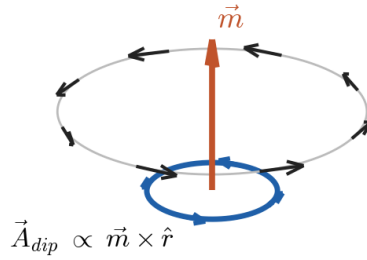
$$\vec{A}_{\text{dip}}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{1}{r^2} I \int d\vec{a}' \times \hat{r} \quad (28)$$

obtained from Stokes' law. Define the **magnetic dipole moment** (current $\times$ area):

$$\vec{m} = I \int d\vec{a}'$$

Magnetic dipole potential

$$\vec{A}_{\text{dip}}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$



$\vec{A}_{\text{dip}}$ circulates around the axis of $\vec{m}$ (azimuthal), just as the current circulates in the loop.

Derivation of (28) from Stokes' theorem

We need $\oint (\hat{r} \cdot \vec{r}') d\vec{l}' = \int d\vec{a}' \times \hat{r}$. Stokes' theorem:

$$\oint \vec{P} \cdot d\vec{l}' = \iint_S (\nabla' \times \vec{P}) \cdot d\vec{a}'$$

Take $\vec{P} = \vec{n} R$, with $\vec{n}$ a constant vector and R a scalar function. Then

$$\iint (\nabla' \times \vec{P}) \cdot d\vec{a}' = - \iint (\vec{n} \times \nabla' R) \cdot d\vec{a}' = \oint \vec{n} \cdot R d\vec{l}' = \vec{n} \cdot \oint R d\vec{l}'$$

because

$$\begin{aligned} \nabla' \times \vec{P} &= \nabla' \times (\vec{n} R) = (\nabla' \times \vec{n}) R - \vec{n} \times (\nabla' R) \\ &= -\vec{n} \times \nabla' R \end{aligned} \quad (29)$$

Using $(\vec{n} \times \nabla' R) \cdot d\vec{a}' = \vec{n} \cdot (\nabla' R \times d\vec{a}')$:

$$\begin{aligned} -\vec{n} \cdot \int (\nabla' R \times d\vec{a}') &= \vec{n} \cdot \oint R d\vec{l}' \\ \Rightarrow - \int (\nabla' R \times d\vec{a}') &= \oint R d\vec{l}' \end{aligned} \quad (30)$$

Now let $R = \hat{r} \cdot \vec{r}'$. Then

$$\nabla' (\hat{r} \cdot \vec{r}') = \hat{r} \times (\nabla' \times \vec{r}') + (\hat{r} \cdot \nabla') \vec{r}' = (\hat{r} \cdot \nabla') \vec{r}' \quad (31)$$

(assignment: check the identity; $\hat{r}$ is constant with respect to the primed coordinates, and $\nabla' \times \vec{r}' = 0$)

Evaluate $(\hat{r} \cdot \nabla')\vec{r}'$ with $\vec{r}' = x'\hat{i} + y'\hat{j} + z'\hat{k}$, $\nabla' = \frac{\partial}{\partial x'}\hat{i} + \frac{\partial}{\partial y'}\hat{j} + \frac{\partial}{\partial z'}\hat{k}$ and $\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \hat{r}_x\hat{i} + \hat{r}_y\hat{j} + \hat{r}_z\hat{k}$:

$$\hat{r} \cdot \nabla' = \hat{r}_x \frac{\partial}{\partial x'} + \hat{r}_y \frac{\partial}{\partial y'} + \hat{r}_z \frac{\partial}{\partial z'} \Rightarrow (\hat{r} \cdot \nabla')\vec{r}' = \hat{r}_x\hat{i} + \hat{r}_y\hat{j} + \hat{r}_z\hat{k} = \hat{r}$$

$$\therefore (\hat{r} \cdot \nabla')\vec{r}' = \hat{r} \quad (32)$$

Putting (32) into (31) and then in (30):

$$-\int \hat{r} \times d\vec{a}' = \int d\vec{a}' \times \hat{r} = \oint (\hat{r} \cdot \vec{r}') d\vec{l}'$$

which is exactly what we needed for (28).

10 Summary quick reference

	Scalar potential V	Vector potential $\vec{A}$
solution (gauge fixed)	$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{\mathcal{R}} d\tau'$	$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{J}}{\mathcal{R}} d\tau'$
monopole ($1/r$)	$\frac{Q}{4\pi\epsilon_0 r}$, scalar	$\propto \oint d\vec{l}' = 0$ (never)
dipole ($1/r^2$)	$\frac{\hat{r} \cdot \vec{p}}{4\pi\epsilon_0 r^2}$, $\vec{p} = \int \vec{r}' \rho d\tau'$	$\frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$, $\vec{m} = I \int d\vec{a}'$
quadrupole ($1/r^3$)	$\frac{1}{4\pi\epsilon_0} \sum_{ij} \frac{Q_{ij} \hat{r}_i \hat{r}_j}{2r^3}$	next term of (26)

Gauge	Condition	Result
Coulomb	$\nabla \cdot \vec{A} = 0$	$\nabla^2 V = -\rho/\epsilon_0$
Lorenz	$\nabla \cdot \vec{A} = -\mu_0\epsilon_0 \partial V/\partial t$	$\square^2 V = -\rho/\epsilon_0$, $\square^2 \vec{A} = -\mu_0 \vec{J}$